Ignytion io

IGNYTION IO · TECHNICAL LAB NOTE

SPI protocol: From RTL to routed GDS

A beginner's walkthrough of a checked open-source reference build

Follow spi_core from its communication rules and Verilog **source** to a routed IHP SG13G2 layout. Worked calculations, measured digital traces and tool results are identified separately. This is an educational block, not a manufactured chip or a signoff-ready design.

STAGE	WHAT IT PRODUCED	RESULT
Simulation	Listed directed checks	PASS directed
Synthesis	131 mapped functional cells	PASS mapping
Routing	Routed DEF and extracted SPEF	PASS reference
Timing	TT and slow setup / hold	PASS reference
GDS	Decoded placement agreement	PASS export
Foundry DRC / LVS	No accepted result	UNPERFORMED

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What is in this document

Start with the wire-level rules, then follow the same circuit through the tools. Each section has a specific evidence boundary.

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SECTION 1 · START HERE

Start here

A protocol is an agreement about how connected devices exchange information: which wire carries each bit, when to sample it, and how to recognise a complete transfer.

WHAT WE BUILT AND WHAT IT PROVES

A Verilog description represents circuits, not software instructions running one after another. Registers remember values; combinational logic calculates the next values. A **clock** advances the registered state. The next pages first explain this protocol at its wires, then connect those rules to the actual reference RTL.

The implemented unit is spi_core in hw/protocols/spi. It is a fixed **reference** block, not the Jane Street programmable combined emulator or a Tiny Tapeout submission top. No new competition architecture, instruction set, pin map or authority decision is implied by these documents.

Only four mode/divider pairs are tested; the hardened core is mode 0, divider 2. General multi-byte CS envelopes are absent. Later pages give the exact stimulus and the implementation evidence rather than using a broad protocol-compliant label. A successful mapped netlist, routed DEF and decoded **GDS** establish concrete implementation/export results, not readiness to manufacture.

The reading path starts with chip-design vocabulary, then develops the protocol, calculations, local interface and failure cases. Separate sections explain lint, simulation, synthesis, placement, routing, timing and **GDS**. The last two pages define design and physical terms. No prior ASIC-flow experience is assumed.

EVIDENCE VOCABULARY

Label	Meaning in this document
Worked example	Explicit assumptions; independently calculated
Measured trace	Positive digital regression VCD, where present
Directed PASS	Listed assertions and negative controls
Routed / decoded GDS	Physical export and structural agreement
Signoff	Not performed or claimed

SECTION 2 · CHIP DESIGN

Chip design in ten minutes

Each stage makes the same circuit more concrete. The folders below are the actual unit layout, not a copied APB command list.

CHIP DESIGN

RTL states what hardware should do. A simulator checks selected input sequences before any silicon exists. Synthesis replaces that description with gates from a process library. Physical tools choose positions and metal routes. Timing analysis checks the resulting delays; stream-out writes the mask-layout geometry.

THE PIPELINE, STAGE BY STAGE

FOLDER	WHAT HAPPENS HERE
00_lint	Read code for risky patterns; retained warnings
01_rtl	Pinned source and the local wrapper
02_sim	Independent stimulus, comparisons and fatal controls
03_verification	Supplemental checks; not a universal UVM PASS
04_synthesis	Map functional logic to SG13G2 cells
05_dft	Manufacturing test work remains unperformed
06_floorplanning	Die outline, rows, power and pin locations
07_placement	Choose legal positions for mapped cells
09_place_and_route	Clock tree, routing and SPEF extraction
08_sta	Read post-route TT and slow timing reports
10_gds	Stream-out and independent placement decoder
11_physical_verification	Foundry DRC / LVS remain unperformed

SECTION 3 · INTERFACE RULES

One clock, two data directions

Controller and peripheral exchange bits simultaneously while one peripheral owns the selected response path.

INTERFACE RULES

The repository keeps conventional MOSI/MISO names. Multiple peripherals generally share **clock** and data, with separate chip-select signals. Only the selected peripheral should drive a shared MISO.

A full-duplex transfer does not double the **clock** count: both wires carry a bit in each period. An implementation may write each bit by index, as this core does, rather than literally shift a register; both can produce the same assembled byte. The outgoing value and incoming value need not match.

Unselected peripherals need appropriate high-impedance behaviour if MISO is shared. Two simultaneously selected push-pull devices can contend, unlike I2C **open-drain** arbitration. A board that uses separate MISO wires has a different topology. The present unit represents one peripheral selection and has no multi-CS decoder; this section explains system wiring rather than claiming that logic already exists.

DIGITAL SIGNAL OWNERSHIP

Signal	Controller / peripheral
SCLK and MOSI	Controller drives
MISO	Selected peripheral drives
CS_n	Controller frames selection
Unselected MISO	Must not contend on a shared wire

SECTION 4 · SAMPLING RULES

Decode CPOL and CPHA as edges

Leading means movement away from idle; trailing means return to idle. It does not always mean rising and falling.

SAMPLING RULES

With CPOL one, leading is falling. These relative names simplify the mode table. The bus should settle to its configured idle polarity before and after a transaction.

For CPHA zero, the first data bit must be available before the first leading edge. For CPHA one, the first leading edge launches data and the trailing edge samples it. Incorrect phase often yields a one-bit shift.

A mode mismatch does not necessarily produce a clean error flag, because SPI has no mandatory acknowledgement or checksum.

FOUR MODE DEFINITIONS

Mode	Idle	Launch/update	Sample
0	Low	Falling, first bit preloaded	Rising
1	Low	Rising	Falling
2	High	Rising, first bit preloaded	Falling
3	High	Falling	Rising

SECTION 5 · TIMING CALCULATION

Divider, latency and return-path margin

The physical reference and the fast simulation have different clock frequencies.

TIMING CALCULATION

With half-bit divider D, the serial frequency is fcore / (2D). The 50 MHz physical **reference** with D=2 therefore has a nominal 12.5 MHz SCLK and a 640 ns eight-bit wire interval. D=3 would give 8.333 MHz and 960 ns, but that is a different elaborated configuration. Handshake and CS overhead add to these wire-only durations.

The bench uses a 100 MHz core, not 50 MHz. Its mode-0 D=2 measured transaction therefore has a 40 ns SCLK period, 25 MHz SCLK and 320 ns across eight periods. These numbers describe the regression only; they do not certify the hardened block at a 100 MHz core **clock**.

A separate board-budget example assumes a 40 ns half-period, 5 ns peripheral **clock**-to-output, 2 ns round-trip delay and 3 ns receiver setup. The remaining 30 ns is arithmetic margin before pad delay, skew and uncertainty. Internal STA closure at the core clock is not proof of the external peripheral timing contract.

SIMULATION PARAMETER COVERAGE

Mode	Divider 2	Divider 3
0	Tested; hardened	Untested
1	Untested	Tested
2	Tested	Untested
3	Untested	Tested

SECTION 6 · RTL INTERFACE

A request pulse, not a stream contract

The interface names resemble ready-valid signalling, but the imported controller expects a one-cycle request.

RTL INTERFACE

spi_core wraps the pinned Nandland SPI_Master. The interface is a launch-pulse contract, not an arbitrary AXI stream. There is no rx_ready input to apply receive backpressure. Upstream ready may initially require clocks after **reset** to settle. Software or a surrounding controller must observe the documented handshake.

The one-cycle delayed valid supports the first-bit preload path. Once the byte has been accepted, the **source** may change its input data because the core holds a copy. However i_TX_DV is a pulse contract, not a general sustained ready-valid stream. Holding it high retriggers the clock-edge setup logic, which is a misuse of this interface.

The entire CS behaviour is this expression, which makes its limited contract easy to inspect. It is not a command sequencer with programmable CS setup/hold or a multi-byte length register. Preserve this boundary in lab exercises.

SOURCE / COMMANDS

```
assign cs_n = !reset_release[1] || (tx_ready && !tx_valid);
```

hw/protocols/spi/01_rtl/spi_core.v:32

SECTION 7 · CONTROL SEQUENCE

Sixteen edges make one byte

The real controller uses an edge counter and generated leading/trailing flags.

CONTROL SEQUENCE

A request loads sixteen edge events for an eight-bit transfer. The divider produces leading and trailing flags; TX and RX processes use those flags according to CPHA. The engine returns ready once no edges remain. Keeping valid asserted reloads the counter, so a normal ready-valid producer cannot be attached without adapting the contract. This is a **source**-derived misuse warning, not a separately measured injected-failure scenario.

The inspected **source** comes directly from the pinned upstream implementation. r_RX_Bit_Count selects the destination bit, so no shift operator is required. o_RX_DV is a completion pulse and has no receive ready signal. A downstream consumer must capture the byte when that pulse appears.

The bench interrupts an active transfer with **reset** and checks recovery. Physical reset integration still requires reset-domain analysis. The core clock should keep running for reset release. Lint reports a mixed synchronous/asynchronous use associated with the synchroniser, which remains documented rather than being hidden.

SOURCE-DERIVED CONTROL SEQUENCE

Phase	State condition	Action
Accept	i_TX_DV pulse	Load 16 edge events, clear ready
Leading edge	Half-period count reached	Toggle clock, pulse leading flag
Trailing edge	Full-period count reached	Toggle clock, pulse trailing flag
Complete	Edge count zero	Return ready, RX valid pulses

SECTION 8 · INTEGRATION LIMITS

A byte envelope is not a flash command

The wrapper deasserts selection after each byte. Many real devices require one continuous envelope around several bytes.

INTEGRATION LIMITS

A hypothetical flash read can use the five-byte sequence 03 00 01 02 00: command 03, a three-byte address 000102, then a dummy transmit byte during which data returns. At 12.5 MHz, forty SCLK periods take 3.2 us before selection and handshake overhead. The device convention must be checked against its data sheet. The current byte wrapper cannot preserve one CS envelope across this complete command.

This is a hypothetical classic three-byte-address read convention; verify the actual flash data sheet before using it. The present wrapper asserts CS for a single byte envelope and does not implement a general multi-byte transaction controller. Five transferred bytes take forty SCLK periods, but keeping CS asserted across them requires another transaction-level design.

The local regression exhausts byte values, which makes it less likely that a lucky test pattern hides a reversal.

One-bit shifts suggest an edge/phase mismatch; all ones can indicate an undriven pulled-up MISO line. Signal-integrity failures may depend on frequency or wiring. These are diagnostic candidates rather than guarantees; inspect the actual peripheral electrical interface and timing.

BEFORE ATTACHING A PERIPHERAL

Question	Required answer
SPI mode and bit order?	From the peripheral data sheet
CS across command / address / data?	Adapt envelope if required
MISO output-disable timing?	Check shared-line contention
Error detection?	Application/device-specific; no mandatory SPI CRC

SECTION 9 · MEASURED TRACE

A5 transmitted, 5A returned

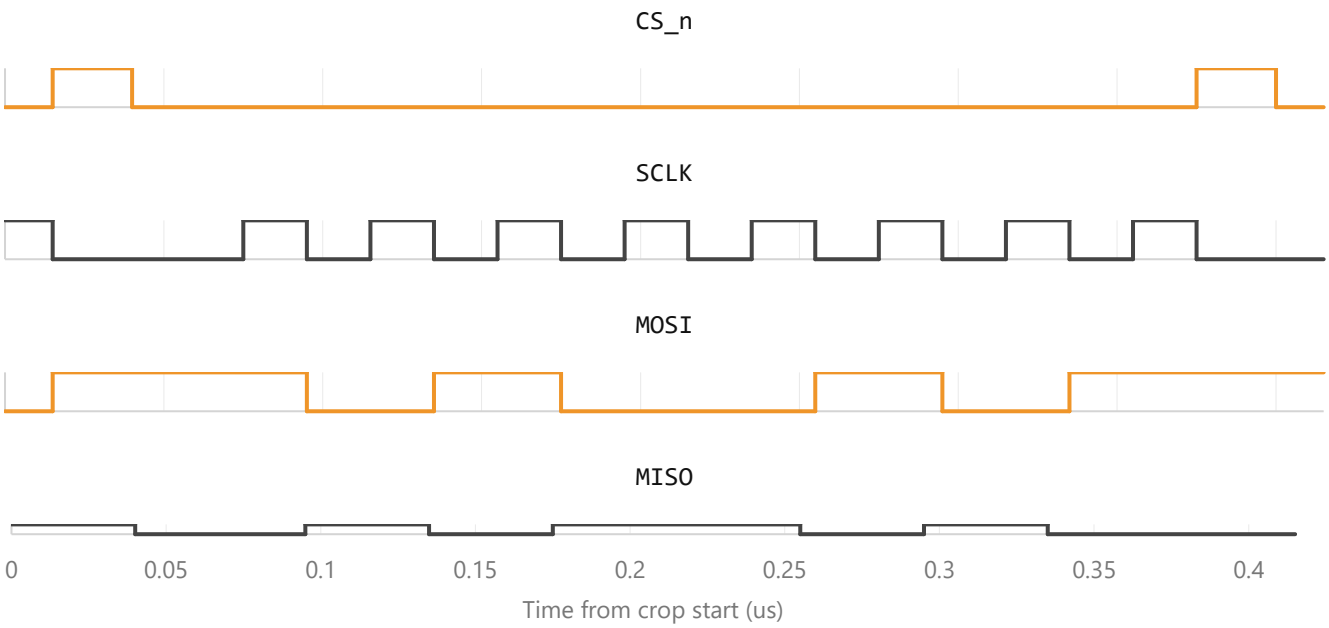
A positive mode-0 transaction is independently decoded in both directions.

MEASURED TRACE

The selected peripheral model returns the complement of each transmitted byte. With MOSI A5, MISO must therefore be 5A. Eight rising SCLK edges inside the active-low CS interval are reconstructed separately for each data wire. Success is not inferred by looping the controller output into its input.

Successive rising edges are 40 ns apart: this trace uses the 100 MHz bench **clock** and divider 2, giving 25 MHz SCLK. The four ideal mode illustrations in the student lesson instead use MISO 3C to demonstrate sampling-edge logic. That deliberate example value is not the responder value measured here. The VCD hash and trace decode remain attached to the figure data.

MEASURED POSITIVE REGRESSION



Logic HIGH=1, LOW=0. Digital simulation, not analogue compliance; source hash retained.

SECTION 10 · LINT

Lint: read code before running it

Verilator checks legal but suspicious code patterns without exercising transactions. A warning count is a review starting point, not proof of correctness.

LINT

The retained report contains 1 warning messages and 0 error messages. Categories present: SYNCASYNCNET. These figures count report message headers; repeated warnings are not necessarily independent defects. The report is hashed in this document evidence manifest.

The invocation deliberately uses -Wno-fatal so warnings remain visible while lint execution can complete. A successful command therefore does not mean every warning was resolved. Width changes, incomplete cases, **reset** use and unconnected ports need review against the wrapper contract and directed tests.

Lint cannot show that a receive byte is correct, that a transaction finishes, or that an external electrical interface is safe. Those questions require a checking testbench, implementation evidence and separate board or signoff work.

RETAINED WARNING REVIEW

MEASURE	VALUE
Warning message headers	1
Error message headers	0
Clean warning-free RTL?	Not claimed
Protocol behaviour proved by lint?	No

SOURCE / COMMANDS

```
make lint
# inspect, do not just count
# 00_lint/lint_report.txt
```

00_lint/Makefile and run_flow.sh: retained Verilator report

SECTION 11 · SIMULATION AND VERIFICATION

Four tested parameter pairs, not eight

A testbench is code that drives the circuit and judges its outputs. These directed checks use independent expected data and deliberately failing controls.

SIMULATION AND VERIFICATION

The bench instantiates mode 0 with divider 2, mode 1 with divider 3, mode 2 with divider 2 and mode 3 with divider 3. Each checks all 256 transmitted byte values against an independently generated complementary receive byte. The mode and divider are correlated; this is not a four-mode by two-divider cross product.

The real hardened wrapper fixes mode 0 and divider 2. Other modes appearing in simulation do not mean that configuration can be changed after fabrication. Likewise, general continuous multi-byte CS framing, multiple peripheral selection and held-valid misuse are not established by the directed byte tests.

A deliberately wrong receive-byte expectation causes a fatal run. Edge and CS checks accompany the data comparison. The checker distinguishes a correct byte from mere end-of-simulation, while still leaving external pad timing and peripheral compatibility outside this proof.

COVERAGE STATEMENT

Property	Status
256 bytes x four listed pairs	Directed PASS
Full mode x divider cross product	Not covered
Hardened mode 0, divider 2	One fixed configuration
Arbitrary peripheral commands / board	Not covered

> **A checker must be able to fail**
>
> Completion alone is insufficient. Read the successful log and the expected fatal rejection together; neither is a protocol-compliance certificate.

SECTION 12 · SYNTHESIS

Synthesis: a description becomes gates

A netlist is a list of logic cells and their connections. Yosys translates the RTL into cells from the IHP SG13G2 library.

SYNTHESIS

A process design kit, or PDK, contains the process-specific files used by implementation tools. This **reference** uses IHP SG13G2, not the SKY130 library used by the golden APB primer. Standard cells are predesigned gates and storage elements with characterised timing and physical views.

The **source** is compiled and checked under the unit workspace before synthesis. Yosys maps the functional logic into the pinned IHP SG13G2 standard-cell library. The current netlist contains 131 functional cells, excluding $scopeinfo metadata. The structural checker rejects remaining generic logic, latches or unresolved instances.

The mapped count excludes $scopeinfo metadata. It is not the count of every placed instance after **clock** buffers, repair cells and fillers are added. It also does not mean that the original protocol has been independently proven for all possible transactions.

MAPPING RESULT

MEASURE	CURRENT RESULT
Top module	spi_core
Mapped functional cells	131
Technology	IHP SG13G2
Unresolved generic logic / latches	Rejected by structural checker

SOURCE / COMMANDS

```
make synth
node verify_build.mjs synth
```

04_synthesis/synth.tcl; verify_build.mjs synthesis checks

SECTION 13 · PHYSICAL IMPLEMENTATION

Floorplan and placement

The floorplan gives the circuit a physical outline. Placement chooses where its cells sit inside that outline.

PHYSICAL IMPLEMENTATION

The die outline for this standalone block is 300 x 300 um. The floorplan establishes rows and **reference** power/pin geometry. The placer must fit cells legally without overlaps and with room for routing. This is a block-level educational layout, not a complete padframe or package design.

Mapped functional logic has 131 cells. After physical additions and fillers, the routed DEF contains 17,622 placed standard-cell instances. A filler occupies empty row sites; it is not another functional logic gate. That distinction explains why the two counts differ so much.

Utilisation is the fraction of core area occupied by placed cell area. It is not simply functional-cell count divided by die size, and it changes between stages. This note does not substitute APB utilisation, row count or I/O count for unavailable protocol-specific measurements.

ACTUAL PHYSICAL REFERENCE

MEASURE	VALUE
Die outline	300 x 300 um
Mapped functional cells	131
Placed standard-cell instances	17,622
Physical inputs	06_floorplanning; 07_placement
Placement agreement	DEF compared with decoded GDS

SECTION 14 · CLOCK AND WIRES

Clock tree and routing

Clock-tree synthesis distributes clocks to registers. Routing draws the metal connections between placed cells.

CLOCK AND WIRES

A register samples its input at a **clock** edge. Clock buffers distribute that edge across the block; the difference in arrival times is called skew. The reference is constrained at 50 MHz / 20 ns.

OpenROAD completes the staged physical flow. The **reference** die is 300 x 300 um. DEF records routed design geometry and cell placement; SPEF captures extracted parasitics used by the timing analysis. The scripts retain stage-specific output so the result can be investigated instead of relying on a final banner.

Real interconnect has resistance and capacitance, called parasitics. Extraction writes these in **SPEF** for post-route timing. The retained router-rule DRC file is empty. That checks the router view of its work; it is not a full foundry-deck check of the exported polygons.

READ THE PHYSICAL EVIDENCE

ARTIFACT	QUESTION IT ANSWERS
09_place_and_route/route.def	Where are cells and routed connections?
09_place_and_route/route.spef	Which parasitics were extracted?
Router-rule DRC report	Any reported routing-rule violations?
08_sta/timing_report.txt	What does extracted reference timing report?

> **Two checks can both be called DRC**
>
> The empty router report is accepted routing evidence. Foundry DRC is a separate unperformed obligation; one result cannot stand in for the other.

SECTION 15 · REFERENCE TIMING

Two measured reference corners

Static timing analysis adds cell and wire delays without simulating input patterns. Setup asks whether data arrives early enough; hold asks whether it stays stable long enough.

REFERENCE TIMING

The physical **clock** constraint is 50 MHz / 20 ns. Post-route timing uses the extracted design rather than an ideal pre-route wire estimate. The report at each corner must be inspected for worst setup and hold slack and for VIOLATED markers. The reported values are not a timing guarantee under unstated board, voltage or temperature conditions.

Current TT setup/hold slack is +15.79 / +0.17 ns. The slow-corner values are +15.73 / +0.33 ns. Both reports contain no VIOLATED marker. These numbers are taken directly from the local report files at document build time; the evidence manifest retains their hashes.

I/O delays are **reference** budgets, not a contract with an external target, transceiver or PHY. Bus synchronisation, reset recovery/removal, all additional corners and modes, signal integrity and board timing need their own analysis. Positive slack in these two reports is therefore called reference timing closure, not foundry or system signoff.

LIVE EXTRACTED SLACK, NS

Corner	Worst setup	Worst hold
TT	15.79	0.17
Slow	15.73	0.33
Status	PASS reference reports	No VIOLATED markers

> **Positive slack has a stated boundary**
>
> Slack is remaining timing margin. These values apply to the listed clocks, I/O budgets and two library corners; they are not a board-timing or manufacturing signoff guarantee.

SECTION 16 · GDS STREAM-OUT

Count the right objects

GDS stores layout polygons and hierarchical cell references. Stream-out combines routed geometry with the real standard-cell layouts from the PDK.

GDS STREAM-OUT

The **GDS** file contains 1,060,628 bytes and 41 structures. The independent decoder finds 17,622 placed standard-cell instances including fillers. That placement count is intentionally much larger than the 131 mapped functional cells: filler instances occupy unused placement sites and structures are unique hierarchy definitions.

The decoder compares exported cell identities, coordinates and orientations against the routed DEF. A deliberately altered placement/export expectation fails, including under Python optimisation, so acceptance is not implemented with removable assert-only checks. Raw file SHA256 is a run-specific fingerprint because stream-out timestamps can change.

The OpenROAD router-rule DRC report is empty. It is not a foundry-deck DRC result. LVS, antenna, scan/DFT, power integrity, padframe, tap/endcap strategy and manufacturing signoff remain unperformed. No empty routing report is promoted into a DRC-clean tapeout claim.

LIVE EXPORT MEASUREMENTS

Quantity	Current value
Mapped functional cells	131
GDS bytes	1,060,628
GDS structures	41
Placed standard-cell instances	17,622
Die	300 x 300 um

SECTION 17 · UNPERFORMED SIGNOFF

DRC and LVS: what is still missing

Design rule checking asks whether geometry follows the process rules. Layout versus schematic asks whether the extracted layout circuit matches the intended netlist.

UNPERFORMED SIGNOFF

Foundry DRC requires the applicable, qualified rule deck and an inspected report on the final polygons. Typical checks include minimum widths, spacing and enclosure. No foundry-deck DRC result has been accepted for this protocol **reference**. The router report is narrower evidence.

LVS requires extracting a circuit from the layout and comparing devices and connectivity with the intended circuit. A structural **GDS** decoder checking cell references, coordinates and orientations does not perform that extraction or connectivity comparison. No accepted LVS match is available.

Antenna, scan/DFT, power integrity, padframe, tap/endcap strategy and manufacturing release remain unperformed. The golden APB primer reports its own separate checks; those results do not transfer to SG13G2 or to this design.

SEPARATE ACCEPTED EXPORT FROM MISSING SIGNOFF

CHECK	THIS REFERENCE
Router-rule report	Empty; narrower routing evidence
Decoded GDS placement	PASS structural/export agreement
Foundry-deck DRC	UNPERFORMED
Extracted-layout LVS	UNPERFORMED
Manufacturing readiness	Not claimed

> **A GDS file is not tapeout approval**
>
> This package is suitable for studying and reproducing a block implementation. Do not hand it off as a signoff-clean manufacturing design.

SECTION 18 · RESULTS SUMMARY

Everything at a glance

Read each result together with the question it actually answers. The same PASS label never implies every protocol or manufacturing obligation is closed.

RESULTS SUMMARY

The directed functional result and the physical export result are both useful, but they prove different things. The physical build uses the fixed wrapper configuration and **reference** constraints. Additional modes, external peripherals, arbitrary clock relationships and board operation require their own evidence.

Important protocol boundary: Only four mode/divider pairs are tested; the hardened core is mode 0, divider 2. General multi-byte CS envelopes are absent.

FULL REFERENCE FLOW

STAGE	TOOL / EVIDENCE	STATUS
Lint	Verilator; 1 retained warnings	Review
Simulation	Independent directed expectations	PASS directed
Negative controls	Deliberate fatal mismatches	PASS rejection
Synthesis	Yosys; 131 mapped cells	PASS mapping
Floorplan / placement	OpenROAD; 300 x 300 um	PASS reference
Routing / extraction	DEF and SPEF retained	PASS reference
Timing	TT and slow setup / hold	PASS reference
GDS	KLayout; 1,060,628 bytes	PASS export
Foundry DRC / LVS	No accepted report	UNPERFORMED
Full-chip / board	No accepted integration proof	UNPERFORMED

SECTION 19 · RUN IT YOURSELF

Run it yourself in WSL

Use the pinned source, the unit tool lock and the external PDK revision together.

RUN IT YOURSELF

The proven entry point is a WSL login Bash shell. igny init prepares the packaging runtime. Unsetting CRUCIBLE_ENV after initialisation allows the unit crucible.toml workspace selection to control tool resolution rather than an inherited environment override. This is the observed workflow, not a change to Crucible internals.

Each unit has its existing environment and lock. The common tool set includes Icarus, Verilator, Yosys, OpenROAD and klayout-stream. The separate PDK checkout is IHP SG13G2 revision 22f43352dd8219f9007eb659e422e0d5fe28c5fb. A tool lock does not pin that external PDK by itself; IHP_PDK_ROOT must point to the matching installation.

The documented machine could not use the pip klayout package as the expected stream-out executable; igny tool install klayout-stream supplies the proven converter. make lint sim checks functional behaviour, make flow executes the staged implementation, and node verify_build.mjs all inspects artifacts. Review normal and intentionally failing logs before accepting the result.

Re-running a simulation can replace its VCD and invalidate an earlier document trace hash. Retain a document evidence manifest or regenerate the measured figures after the positive run. Do not silently preserve an old chart while claiming it is the newly generated waveform.

SOURCE / COMMANDS

```
igny init
unset CRUCIBLE_ENV
cd hw/protocols/spi
igny workspace sync
igny tool install klayout-stream
make lint sim
make flow
node verify_build.mjs all
```

Run after entering the repository in WSL login Bash; PDK required.

SECTION 20 · SOURCES AND REMAINING WORK

Sources, ownership and remaining work

The pinned revision identifies the baseline; local integration changes are described separately.

SOURCES AND REMAINING WORK

Upstream: https://github.com/nandland/spi-master
Pinned commit: 1ab581969f8ea2f794072a144b55cd74e5a54f34
Licence: MIT, russell-merrick, 2019. Original **source** notices remain in the RTL. The document is authored by Ignytion; this does not transfer ownership of upstream code or remove its licence obligations.

Local modifications, directed test stimulus and **source** boundaries are recorded in hw/protocols/spi/README.md and the repository append-only .context.md. The first three protocol lessons have separate editable 16:9 student decks; this A4 technical note is a reading and evidence document, not a replacement for their instructor notes.

Primary reading: Microchip TB3215 for SPI edge conventions, plus the pinned Nandland implementation.
Primary guide:
https://www.microchip.com/content/dam/mchp/documents/MCU08/ApplicationNotes/ApplicationNotes/TB3215-Getting-Started-with-SPI-DS90003215.pdf

Open work remains explicit: Only four mode/divider pairs are tested; the hardened core is mode 0, divider 2. General multi-byte CS envelopes are absent. Full-chip integration, manufacturing checks and board validation are not closed by the standalone **reference** flow. The reader can reproduce the listed evidence without assuming those larger obligations have already been completed.

SECTION 21 · DESIGN VOCABULARY

Glossary: design and protocol

Definitions for reading the protocol rules and the checked RTL. Physical implementation terms follow on the next page.

DESIGN AND VERIFICATION

TERM	WHAT IT MEANS
Protocol	Rules for exchanging information between devices
RTL	Registers and the logic moving values between them
Verilog	A language used to describe hardware
FSM	A circuit progressing through defined states
Lint	Static source checks; no transaction execution
Testbench	Code that supplies inputs and checks outputs
Coverage	Which defined situations were exercised
Negative control	An intentionally wrong case the checker must reject
VCD	Digital signal-change trace from a simulation
Handshake	Signals agreeing when data is accepted
CPOL	Clock idle polarity
CPHA	Leading or trailing sample-edge selection
CS_n	Active-low peripheral selection
Half-bit divider	Core cycles between serial clock transitions

SECTION 22 · PHYSICAL VOCABULARY

Glossary: implementation and signoff

These definitions help distinguish a successfully exported reference layout from a manufacturing-ready design.

PHYSICAL IMPLEMENTATION

TERM	WHAT IT MEANS
Synthesis	Translate RTL into a technology-cell circuit
Netlist	Cells and the connections between them
Standard cell	Predesigned logic/storage cell in a process library
PDK	Process design kit: process-specific implementation files
SG13G2	IHP process used by these reference builds
Floorplan	Die/core outline, rows, power and pin geometry
Placement	Choose legal positions for cells
Filler	Physical row-site cell, not functional logic
CTS / skew	Clock distribution / difference in arrival times
Routing / via	Metal connections / links between metal layers
Parasitics / SPEF	Wire resistance/capacitance / extracted data format
Corner	A characterised process, voltage and temperature case
STA / slack	Static timing analysis / remaining timing margin
DEF / GDS	Placement/routing exchange / polygon hierarchy
DRC	Check geometry against applicable process rules
LVS	Compare the extracted layout circuit with its netlist
Signoff	Accepted applicable release checks; unperformed here

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